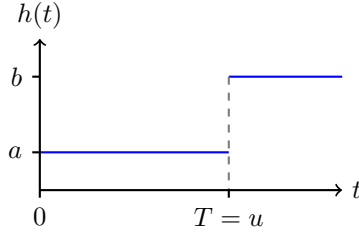


Q. 1 (Heart attack). Let us first write down the precise meaning of our modelling assumption: we assume that the *conditional* hazard rate of the lifetime L given $T = u$ is given by

$$h(t) = \begin{cases} a & \text{if } t < u, \\ b & \text{if } t \geq u. \end{cases}$$



In other words,

$$\begin{aligned} \mathbb{P}\{L > t | T = u\} &= e^{-\int_0^t h(s) ds} = \begin{cases} e^{-\int_0^t a ds} & \text{if } t < u \\ e^{-(\int_0^u a ds + \int_u^t b ds)} & \text{if } t \geq u \end{cases} \\ &= \begin{cases} e^{-at} & \text{if } t < u \\ e^{-(au + b(t-u))} & \text{if } t \geq u. \end{cases} \end{aligned}$$

a. We first compute the conditional expectation

$$\begin{aligned} \mathbb{E}(L | T = u) &= \int_0^\infty \mathbb{P}\{L > t | T = u\} dt \\ &= \int_0^u e^{-at} dt + \int_u^\infty e^{-au - b(t-u)} dt \\ &= \frac{1}{a} \int_0^u a e^{-at} dt + e^{-au} \int_0^\infty e^{-bt} dt \\ &= \frac{1 - e^{-au}}{a} + \frac{e^{-au}}{b} \\ &= \frac{1}{a} - \frac{(b-a)}{ab} e^{-au}. \end{aligned}$$

Hence, the expected lifetime is

$$\begin{aligned} \mathbb{E}(L) &= \int_0^\infty \mathbb{E}(L | T = u) \mathbb{P}\{T \in du\} \\ &= \int_0^\infty \left(\frac{1}{a} - \frac{(b-a)}{ab} e^{-au} \right) \mu e^{-\mu u} du \\ &= \frac{1}{a} \int_0^\infty \mu e^{-\mu u} du - \frac{\mu(b-a)}{ab} \int_0^\infty e^{-(a+\mu)u} du \\ &= \frac{1}{a} - \frac{\mu(b-a)}{ab} \cdot \frac{1}{a+\mu} \\ &= \frac{b+\mu}{b(a+\mu)}. \end{aligned}$$

b. The required probability is given by

$$\begin{aligned}
\mathbb{P}\{L < T\} &= \int_0^\infty \mathbb{P}\{L < u | T = u\} \mu e^{-\mu u} du \\
&= \int_0^\infty (1 - \mathbb{P}\{L > u | T = u\}) \mu e^{-\mu u} du \\
&= \int_0^\infty (1 - e^{-au}) \mu e^{-\mu u} du \\
&= 1 - \frac{\mu}{a + \mu} \\
&= \frac{a}{a + \mu}.
\end{aligned}$$

Q. 2. Let X be the non-smoker's lifespan (with hazard rate function h) and let Y be the smoker's lifespan (with hazard rate function $2h$). They are independent, and their distributions are given by

$$\begin{aligned}
\mathbb{P}\{X > t\} &= \exp\left(-\int_0^t h(u) du\right) \\
\mathbb{P}\{Y > t\} &= \exp\left(-\int_0^t 2h(u) du\right)
\end{aligned}$$

where h satisfies $\int_0^\infty h(u) du = \infty$.

We need to find

$$\mathbb{P}\{Y > X | X > 50, Y > 50\}.$$

Conditioned on $\{X > 50, Y > 50\}$, X and Y are still independent. Their conditional distributions are given by (for $t > 50$):

$$\begin{aligned}
\mathbb{P}\{X > t | X > 50, Y > 50\} &= \mathbb{P}\{X > t | X > 50\} \\
&= \frac{\mathbb{P}\{X > t\}}{\mathbb{P}\{X > 50\}} \\
&= \frac{e^{-\int_0^t h(u) du}}{e^{-\int_0^{50} h(u) du}} \\
&= \exp\left(-\int_{50}^t h(u) du\right).
\end{aligned}$$

Similarly

$$\mathbb{P}\{Y > t | X > 50, Y > 50\} = \exp\left(-\int_{50}^t 2h(u) du\right).$$

By taking derivative, we can find the conditional density of X :

$$\mathbb{P}\{X \in dt | X > 50, Y > 50\} = h(t) \exp\left(-\int_{50}^t h(u) du\right) dt, \quad t > 50.$$

Now we can compute the probability by conditioning on T :

$$\begin{aligned}
\mathbb{P}\{Y > X | X > 50, Y > 50\} &= \int_{50}^{\infty} \mathbb{P}\{Y > t | X > 50, Y > 50, X = t\} \mathbb{P}\{X \in dt | X > 50, Y > 50\} \\
&= \int_{50}^{\infty} \mathbb{P}\{Y > t | Y > 50\} \mathbb{P}\{X \in dt | X > 50, Y > 50\} \\
&= \int_{50}^{\infty} \exp\left(-\int_{50}^t 2h(u)du\right) h(t) \exp\left(-\int_{50}^t h(u)du\right) dt \\
&= \int_{50}^{\infty} h(t) \exp\left(-\int_{50}^t 3h(u)du\right) dt \\
&= -\frac{1}{3} \exp\left(-\int_{50}^t 3h(u)du\right) \Big|_{50}^{\infty} = \boxed{\frac{1}{3}}.
\end{aligned}$$

Q. 3 (Superposition and thinning). We are given independent Poisson processes $\{N_t^1\}, \dots, \{N_t^k\}$ with rates $\mu_1, \dots, \mu_k$, respectively, and we let $N_t = N_t^1 + N_t^2 + \dots + N_t^k$.

- a. By the superposition principle, we know that $\{N_t\}$ is a Poisson process with rate $\mu = \mu_1 + \dots + \mu_k$. In particular, for fixed time t , N_t has the Poisson distribution with parameter μt , in other words,

$$\mathbb{P}\{N_t = n\} = \frac{e^{-\mu t} (\mu t)^n}{n!}, \quad n \in \{0, 1, 2, \dots\}.$$

- b. We write $X_\ell = i$ if the ℓ th arrival of $\{N_t\}$ is an arrival of $\{N_t^i\}$. In other words, X_ℓ is the label of the ℓ th arrival of $\{N_t\}$. To find the distribution of X_ℓ , recall how we apply the thinning principle for $\{N_t\}$ to recover the processes $\{N_t^1\}, \dots, \{N_t^k\}$. We label the arrivals of $\{N_t\}$ with labels $1, 2, \dots, k$ independent of each other and of $\{N_t\}$. We assign the probability that an arrival has the i th label as μ_i/μ so that the arrivals with the i th label form a Poisson process with rate $\mu \cdot \frac{\mu_i}{\mu} = \mu_i$ by thinning! The thinning principle also implies that the resulting processes corresponding to each of the labels are independent. Therefore, the arrivals with the i th label is $\{N_t^i\}$ (or a copy of it with the same properties).

Now that we obtain $\{N_t^i\}$ as a result of thinning, $\mathbb{P}\{X_\ell = i\}$ is the probability that the ℓ th arrival of $\{N_t\}$ has the i th label. Hence,

$$\mathbb{P}\{X_\ell = i\} = \frac{\mu_i}{\mu}, \quad i \in \{1, \dots, k\}.$$

- c. As a result of the reconstruction in a., $\mathbb{P}\{X_1 = i, X_2 = j\}$ is the probability that the first arrival of $\{N_t\}$ has the i th label and the second arrival has the j th label. Since labels are assigned independently,

$$\mathbb{P}\{X_1 = i, X_2 = j\} = \frac{\mu_i \mu_j}{\mu^2} = \mathbb{P}\{X_1 = i\} \mathbb{P}\{X_2 = j\}, \quad i, j \in \{1, \dots, k\},$$

which gives the joint distribution of X_1, X_2 and that X_1, X_2 are independent.

Q. 4 (Poisson gymnastics). We are given a Poisson process $\{N_t\}$ with rate λ .

- a. By the definition of conditional probability,

$$\mathbb{P}\{N_4 = 5 | N_9 = 13\} = \frac{\mathbb{P}\{N_4 = 5, N_9 = 13\}}{\mathbb{P}\{N_9 = 13\}}.$$

Note that N_4 and N_9 are *not* independent. However, N_4 and $N_9 - N_4$ are independent by the independent increments property of the Poisson process $\{N_t\}$. Observing that

$$\{N_4 = 5, N_9 = 13\} = \{N_4 = 5, N_9 - N_4 = 8\},$$

we compute the desired probability by

$$\begin{aligned}
\mathbb{P}\{N_4 = 5 | N_9 = 13\} &= \frac{\mathbb{P}\{N_4 = 5, N_9 - N_4 = 8\}}{\mathbb{P}\{N_9 = 13\}} \\
&= \frac{\mathbb{P}\{N_4 = 5\} \mathbb{P}\{N_9 - N_4 = 8\}}{\mathbb{P}\{N_9 = 13\}} \\
&= \frac{\frac{e^{-4\lambda}(4\lambda)^5}{5!} \cdot \frac{e^{-5\lambda}(5\lambda)^8}{8!}}{\frac{e^{-9\lambda}(9\lambda)^{13}}{13!}} \\
&= \frac{13!}{5! \cdot 8!} \frac{4^5 \cdot 5^8}{9^{13}} \\
&= \binom{13}{5} \left(\frac{4}{9}\right)^5 \left(\frac{5}{9}\right)^8,
\end{aligned}$$

where we use the stationary increments property to get the third equality. Does this probability look familiar?

b. Let us distinguish the cases $t > s$, $t = s$ and $t < s$.

First, let us consider the case $t > s$. Since the Poisson process is nondecreasing as a counting process, we have

$$\mathbb{P}\{N_t = k | N_s = n\} = 0 \quad k \in \{0, 1, \dots, n-1\}.$$

Now let $k \geq n$. Note that N_s and $N_t - N_s$ are independent increments of the Poisson process. Hence,

$$\begin{aligned}
\mathbb{P}\{N_t = k | N_s = n\} &= \mathbb{P}\{N_t - N_s + N_s = k | N_s = n\} \\
&= \mathbb{P}\{N_t - N_s + n = k | N_s = n\} \\
&= \mathbb{P}\{N_t - N_s = k - n\} \\
&= \frac{e^{-\lambda(t-s)}(\lambda(t-s))^{k-n}}{(k-n)!}.
\end{aligned}$$

Therefore, given $N_s = n$, the conditional distribution of N_t is the same as the distribution of $n + M$, where $M \sim \text{Pois}(\lambda(t-s))$.

In the case $t = s$, we have $\mathbb{P}\{N_t = k | N_s = n\} = 1$ when $k = n$, and 0 otherwise.

Next, we consider the case $t < s$. Using the fact that the process is nondecreasing, we have

$$\mathbb{P}\{N_t = k | N_s = n\} = 0, \quad k \in \{n+1, n+2, \dots\}.$$

Let $k \in \{0, \dots, n\}$. By the same ideas as in part a., we have

$$\begin{aligned}
\mathbb{P}\{N_t = k | N_s = n\} &= \frac{\mathbb{P}\{N_t = k, N_s = n\}}{\mathbb{P}\{N_s = n\}} \\
&= \frac{\mathbb{P}\{N_t = k, N_s - N_t = n - k\}}{\mathbb{P}\{N_s = n\}} \\
&= \frac{\mathbb{P}\{N_t = k\} \mathbb{P}\{N_s - N_t = n - k\}}{\mathbb{P}\{N_s = n\}} \\
&= \frac{\frac{e^{-\lambda t}(\lambda t)^k}{k!} \cdot \frac{e^{-\lambda(s-t)}(\lambda(s-t))^{n-k}}{(n-k)!}}{\frac{e^{-\lambda s}(\lambda s)^n}{n!}} \\
&= \frac{n!}{k! \cdot (n-k)!} \frac{t^k \cdot (s-t)^{n-k}}{s^n} \\
&= \binom{n}{k} \left(\frac{t}{s}\right)^k \left(\frac{s-t}{s}\right)^{n-k},
\end{aligned}$$

the third equality following from independence of the increments N_t and $N_s - N_t$ for $t < s$.

Thus, given $N_s = n$, the conditional distribution of N_t is $\text{Binom}(n, \frac{t}{s})$.

- c. We suppose that each arrival of $\{N_t\}$ is a man or a woman with probability $\frac{1}{2}$. The genders of the arrivals are independent of each other and of the arrival times. Let M_t be the number of women that arrived by time t . By the thinning principle, $\{M_t\}$ (women) and $\{N_t - M_t\}$ (men) are independent Poisson processes with the same rate $\frac{\lambda}{2}$. In particular, M_t and $N_t - M_t$ are independent. Hence,

$$\begin{aligned}\mathbb{E}(N_t | M_t = m) &= \mathbb{E}(M_t + (N_t - M_t) | M_t = m) \\ &= \mathbb{E}(m + (N_t - M_t) | M_t = m) \\ &= m + \mathbb{E}(N_t - M_t) \\ &= m + \frac{\lambda}{2} \cdot t.\end{aligned}$$

Q. 5 (Buttons). We let X_k be the color of the k th button pressed by the monkey. It is assumed that $\{X_k\}$ are independent and $\mathbb{P}\{X_k = \text{red}\} = \frac{1}{3}$, $\mathbb{P}\{X_k = \text{green}\} = \frac{2}{3}$.

- a. We let R_n be the number of times the monkey pressed the red button after pressing n buttons in total, and G_n the number of times the monkey pressed the green button. Note that the sum $R_n + G_n = n$ is nonrandom. Hence,

$$\mathbb{P}\{R_n = k, G_n = \ell\} = 0 \text{ if } k + \ell \neq n.$$

On the other hand, note that

$$\{R_n = k, G_n = n - k\} = \{R_n = k\}.$$

Since each pressed button is red with probability $\frac{1}{3}$, independent of the other buttons, we have $R_n \sim \text{Binom}(n, \frac{1}{3})$. So

$$\mathbb{P}\{R_n = k, G_n = n - k\} = \mathbb{P}\{R_n = k\} = \binom{n}{k} \left(\frac{1}{3}\right)^k \left(\frac{2}{3}\right)^{n-k}, \quad k \in \{0, \dots, n\}.$$

Note that R_n and G_n are *not* independent. Indeed, knowing the value of R_n completely determines the value of G_n :

$$\mathbb{P}\{G_n = n - k | R_n = k\} = 1 \neq \binom{n}{k} \left(\frac{1}{3}\right)^k \left(\frac{2}{3}\right)^{n-k} = \mathbb{P}\{R_n = k\} = \mathbb{P}\{G_n = n - k\}.$$

- b. We are given a Poisson random variable N with mean a , independent of $\{X_k\}$. One way of computing the joint distribution of R_N and G_N is to observe that

$$R_N + G_N = N,$$

and hence

$$\mathbb{P}\{R_N = k, G_N = \ell\} = \mathbb{P}\{R_N = k, N = k + \ell\} = \mathbb{P}\{R_{k+\ell} = k\} \mathbb{P}\{N = k + \ell\}.$$

The rest of the computation is left as an exercise.

Here is an alternative approach using the thinning principle. Let us consider N as the value of a Poisson process at time $t = 1$: we artificially create a Poisson arrival process $\{M_t\}$ with rate a so that $M_1 = N$. We also suppose that $\{M_t\}$ and $\{X_k\}$ are independent. (There is no harm in assuming this as we are already given that N and $\{X_k\}$ are independent.) Next, we label by X_k the k th arrival of $\{M_t\}$. So each arrival, independent of the others and of the arrival times, is labeled as red with probability $\frac{1}{3}$ and as green with

probability $\frac{2}{3}$. Then, R_{M_t} is the number of arrivals by time t that are labeled as red and G_{M_t} is the number of arrivals by time t that are labeled as green.

By the thinning principle $\{R_{M_t}\}$ and $\{G_{M_t}\}$ are independent Poisson processes with rates $\frac{a}{3}$, $\frac{2a}{3}$, respectively. In particular, $R_{M_1} = R_N$ and $G_{M_1} = G_N$ are independent Poisson random variables with parameters $\frac{a}{3}$, $\frac{2a}{3}$, respectively.